$$f(e) = e_0 + \sum_{e=1}^{\infty} e\left(Adress + Promotion + ecommerce\right)$$

ኢጵላሣ

EplusApp

e^{+up}

ጵጵጵ

888

ኢጵላሣጵ

የዲጂታል መረጃ ምጥቀቶችና የአክሲዮን ገበያ

ተቋም

EplusApp

Software Developmenet, Stock Market & e-Commerce Company

eplusapp.et

$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

e^{+u_p}

ጵ

ጵጵጵ

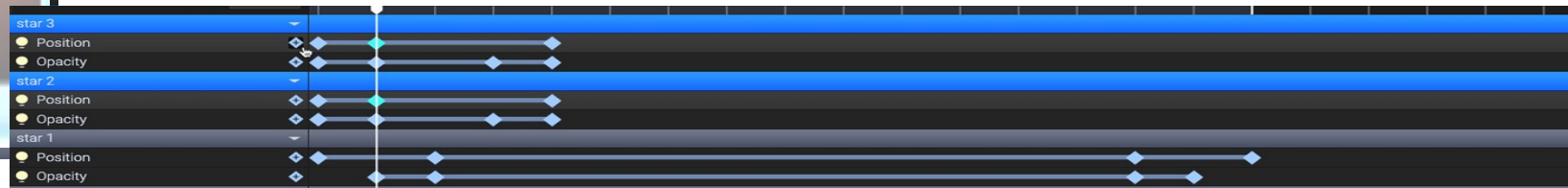
888

Software Development Life Cycle(SDLC)

What Is ?

software development life cycle (SDLC) is a framework that defines the steps involved in the development of software at each phase.

It covers the detailed plan for building, deploying and maintaining the software. SDLC defines the complete cycle of development i.e. all the tasks involved in planning, creating, testing, and deploying a Software Product.



$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

ጵ

ጵጵጵ

888

e^{+u_p}

Software Development Life Cycle(SDLC)

1, Requirement Gathering and Analysis

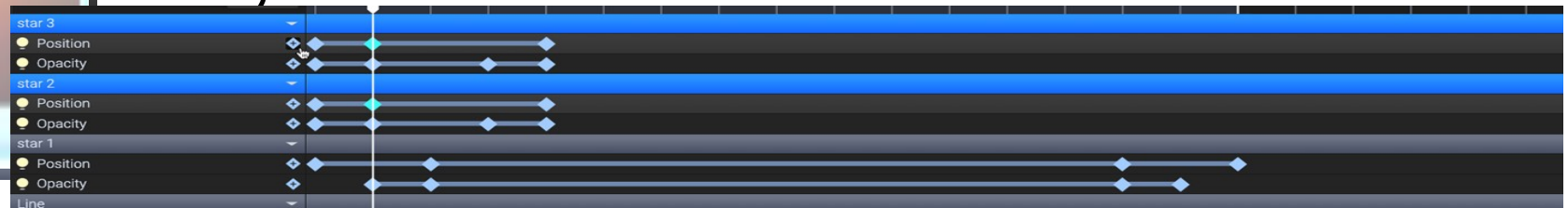
2, Design

3, Implementation or Coding

4, Testing

5,Deployment

6,maintenance



$$f(e) = e_0 + \sum_{e=1}^{\infty} e\left(Adress + Promotion + ecommerce\right)$$

ኢጵላሣ

EplusApp

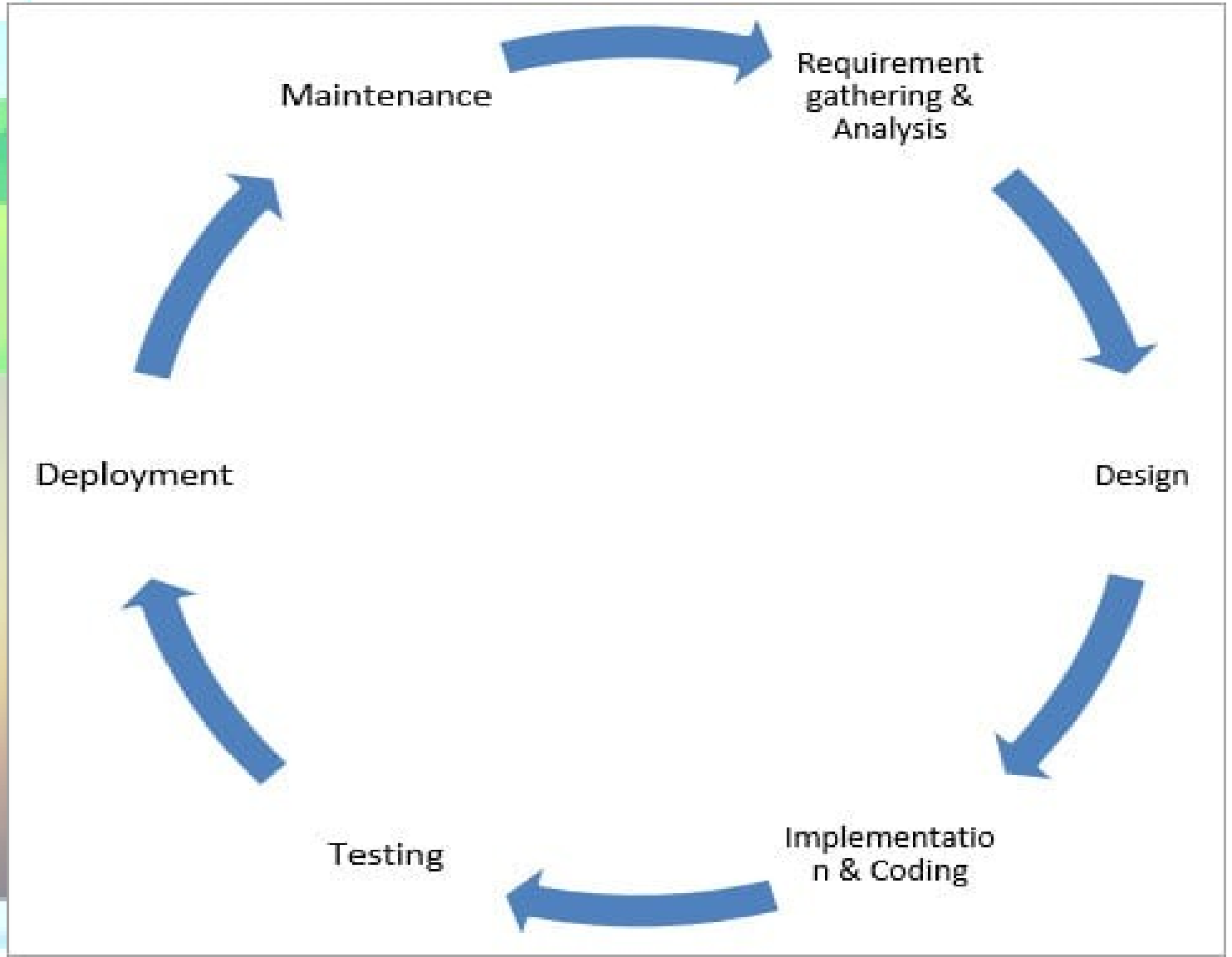
e^{+u_p}

ጵ

ጵጵጵ

888

Software Development Life Cycle(SDLC)



$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

e^{+u_p}

ጵ

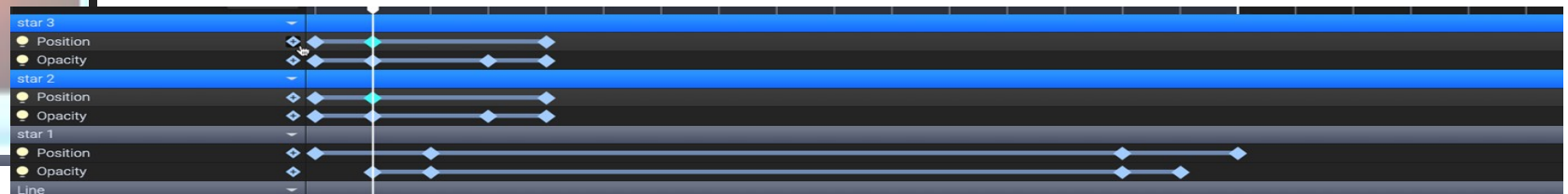
ጵጵጵ

888

Software Development Life Cycle(SDLC)

1, Requirement Gathering and Analysis

- All the relevant information is collected
- Any ambiguities must be resolved
- Business analyst and Project Manager set up a meeting with the customer to gather all the information
- Feasibility study
- SRS Document created



$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

e^{+u_p}

ጵ

ጵጵጵ

888

Software Development Life Cycle(SDLC)

2, Design

- SRS document is used as an input and software architecture that is used for implementing system development is derived.

3, Implementation or Coding

- starts once the developer gets the Design document.
- The Software design is translated into source code.
- All the components of the software are implemented in this phase.

$$f(e)=e_0+\sum_{e=1}^{\infty} e(Adress+Promotion+ecommerce)$$

ኢጵላሣ

EplusApp

e^{+u_p}

ጵ

ጵጵጵ

888

Software Development Life Cycle(SDLC)

4, Testing

- Testing starts once the coding is complete and the modules are released for testing.
- The developed software is tested thoroughly and any defects found are assigned to developers to get them fixed.
- Retesting, regression testing is done until the point at which the software is as per the customer’s expectation.
- Testers refer SRS document to make sure that the software is as per the customer’s standard.

$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

e^{+u_p}

ጵ

ጵጵጵ

888

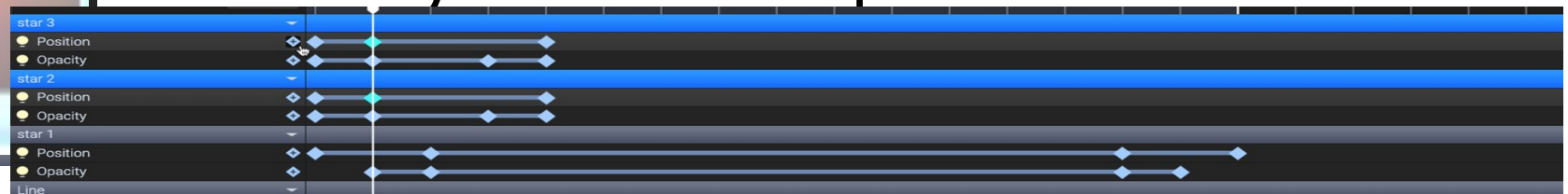
Software Development Life Cycle(SDLC)

5,Deployment

- Once the product is tested, it is deployed in the production environment

6,Maintenance

- After the deployment,maintenance of the product i.e. if any issue comes up and needs to be fixed or any enhancement is to be done is taken care by the developers.



$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

ጵ

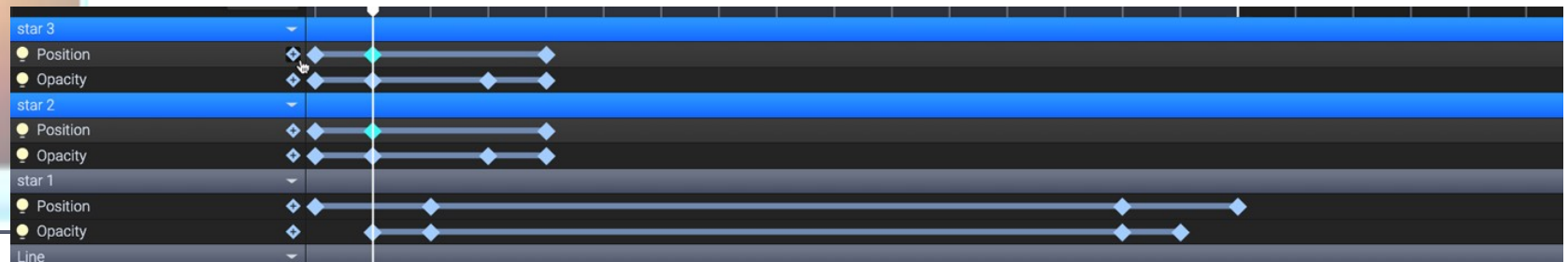
ጵጵጵ

888

e^{+u_p}

Software Development Life Cycle Models

- A software life cycle model is a descriptive representation of the software development cycle.
- SDLC models might have a different approach but the basic phases and activity remain the same for all the models.



$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

ጵ

ጵጵጵ

888

e^{+u_p}

1) Waterfall Model

- The outcome of one phase is the input for the next phase.
- Is the very first model that is used in SDLC. It is also known as the linear sequential model.
- SDevelopment of the next phase starts only when the previous phase is complete.

$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

ጵ

ጵጵጵ

888

e^{+u_p}

1) Waterfall Model

$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ
EplusApp
ጵ
ጵጵጵ
888

e^{+u_p}

1) Waterfall Model

Advantages:

- Easy to understand
- No any complexity

Disadvantages

- Time-consuming
- Cannot be used for the projects which have uncertain requirement
- Any requirement change in the later stages would lead to cost higher as the changes would be required in all the phases.

$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

ጵ

ጵጵጵ

888

e^{+u_p}

2) V-Shaped Model

- V-Model is also known as Verification and Validation Model.
- In this model Verification & Validation goes hand in hand i.e. development and testing goes parallel.
- V model and waterfall model are the same except that the test planning and testing start at an early stage in V-Model.

$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

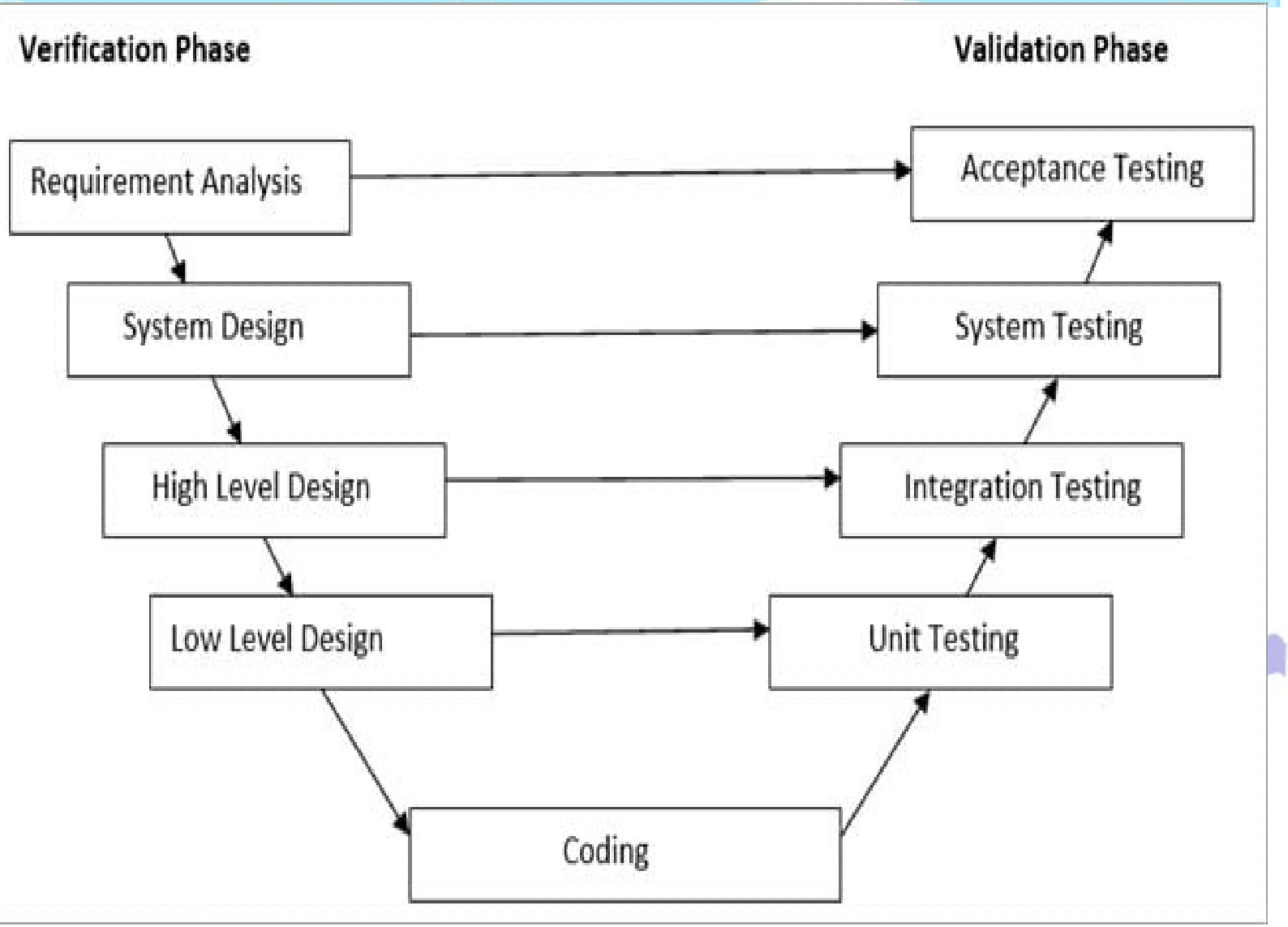
ጵ

ጵጵጵ

888

e^{+u_p}

2) V-Shaped Model



$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢ ጽ ላሣ ጽ

ኢጽላሣ

EplusApp

e^{+u_p}

ጽ

ጽጽጽ

888

Advantages of V – Model

- It is a simple and easily understandable model.
- V –model approach is good for smaller projects wherein the requirement is defined and it freezes in the early stage.
- It is a systematic and disciplined model which results in a high-quality product.

$$f(e) = e_0 + \sum_{e=1} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

ጵ

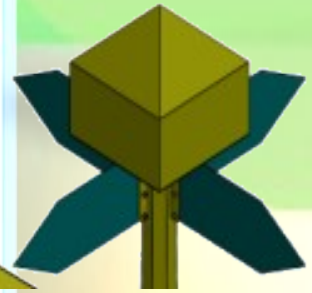
ጵጵጵ

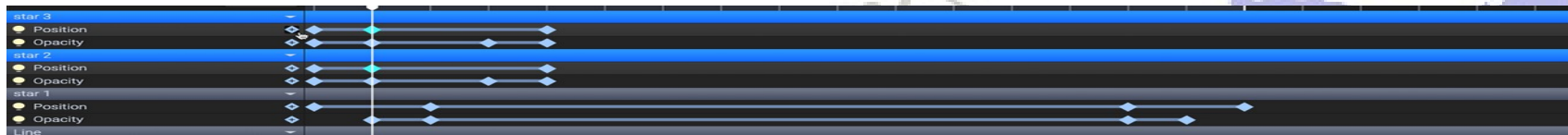
888

e^{+up}

Disadvantages of V-Model:

- V-shaped model is not good for ongoing projects.
- Requirement change at the later stage would cost too high.





$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

ጵ

ጵጵጵ

888

e^{+u_p}

3) Prototype Model

- The prototype model is a model in which the prototype is developed prior to the actual software to get valuable feedback from the customer.
- Feedbacks are implemented and the prototype is again reviewed by the customer for any change.
- This process goes on until the model is accepted by the customer.

$$f(e) = e_0 + \sum_{e=1} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

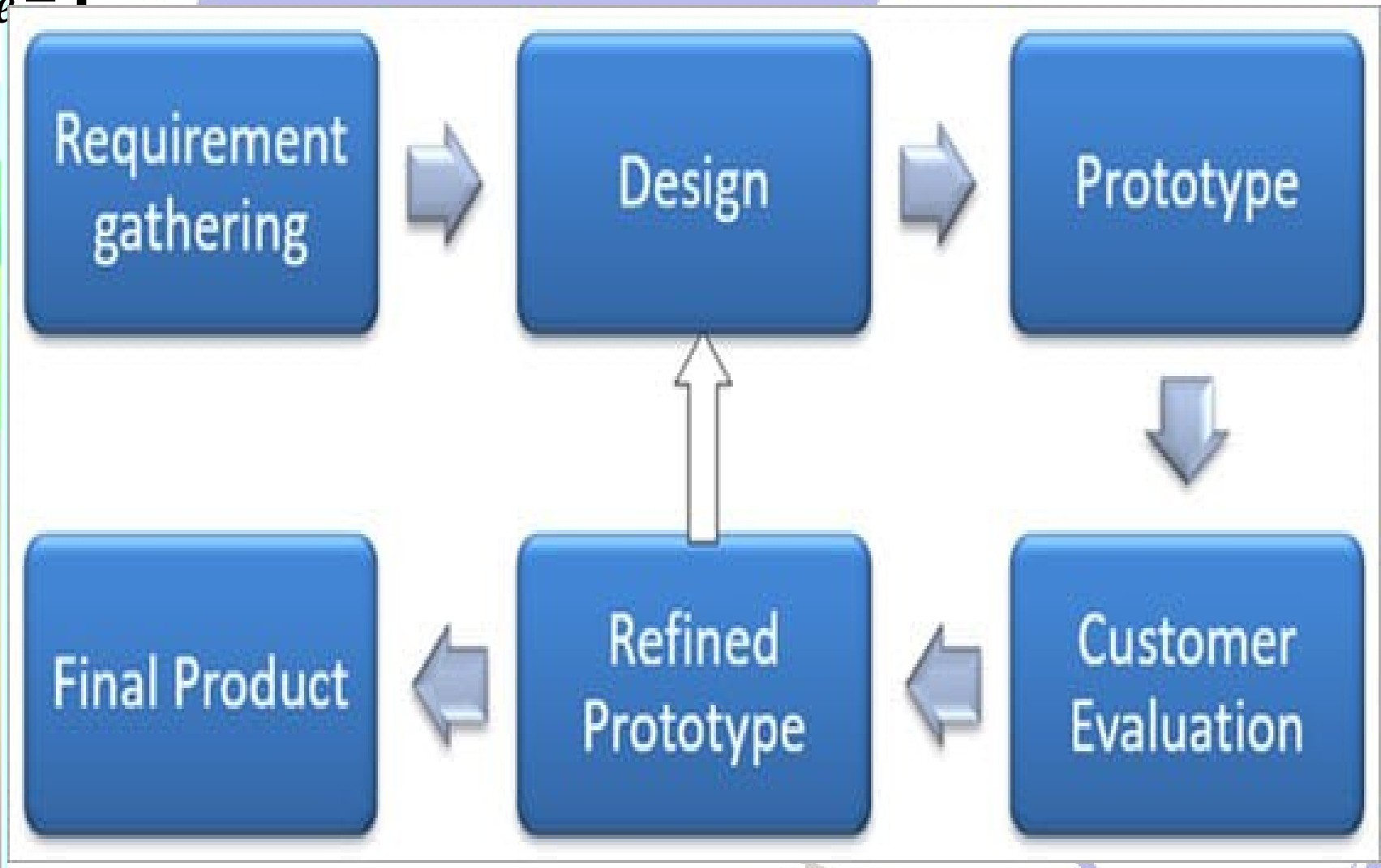
ጵ

ጵጵጵ

888

e^{+u_p}

3) Prototype Model



$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

ጵ

ጵጵጵ

888

e^{+up}

3) Prototype Model

- The actual software is build using the Waterfall model approach.

Advantages of Prototype Model:

- Prototype model reduces the cost and time of development as the defects are found much earlier.
- Missing feature or functionality or a change in requirement can be identified in the evaluation phase and can be implemented in the refined prototype.
- Involvement of a customer from the initial stage reduces any confusion in the requirement or understanding of any functionality.

$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

ጵ

ጵጵጵ

888

e^{+up}

3) Prototype Model

Disadvantages of Prototype Model:

- Since the customer is involved in every phase, the customer can change the requirement of the end product which increases the complexity of the scope and may increase the delivery time of the product.

$$f(e) = e_0 + \sum_{e=1} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

e^{+up}

ጵ

ጵጵጵ

888

4) Spiral Model

- Spiral Model:includes iterative and prototype approach.
- Spiral model phases are followed in the iterations.
- The loops in the model represent the phase of the SDLC process.

  i.e. the innermost loop is of requirement gathering & analysis which follows the Planning, Risk analysis, development, and evaluation.
- Next loop is Designing followed by Implementation & then testing.

  Spiral Model has four phases:

1) Planning
2) Risk Analysis
3) Engineering
4) Evaluation

$$f(e) = e_0 + \sum_{e} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

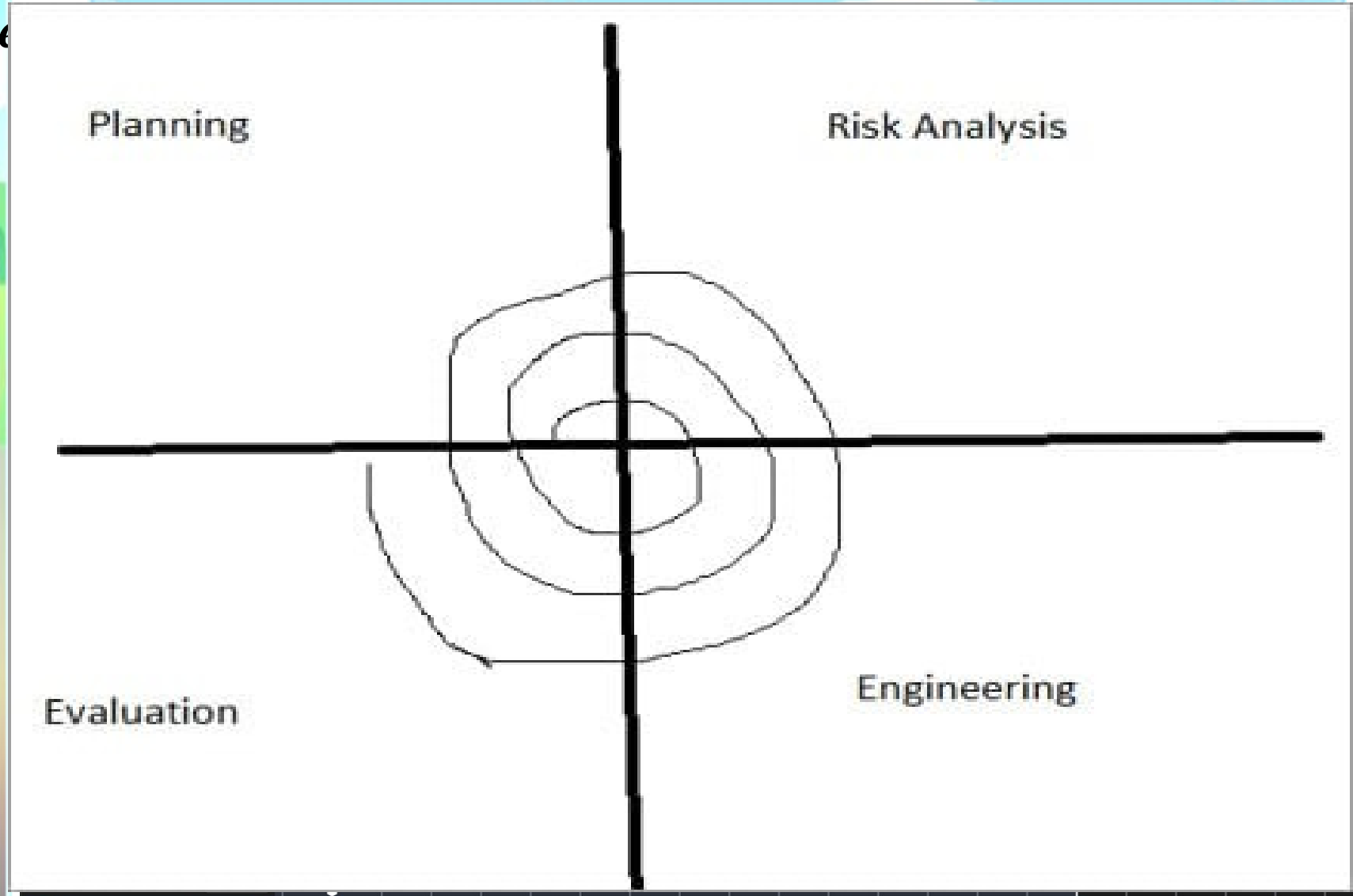
e^{+u_p}

ጵ

ጵጵጵ

888

4) Spiral Model



$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

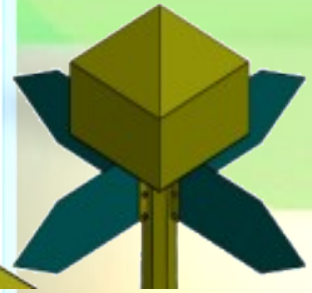
ጵ

ጵጵጵ

888

e^{+up}

4) Spiral Model

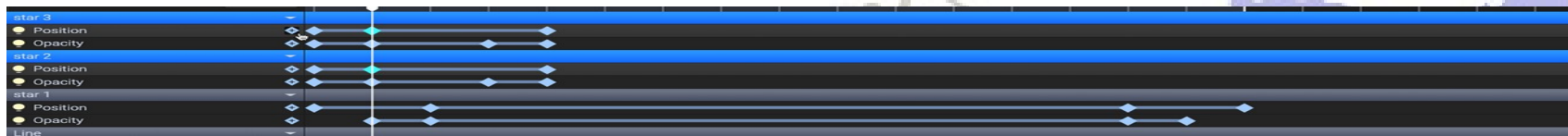


Advantages of Spiral Model:

- Risk Analysis is done extensively using the prototype models.
- Any enhancement or change in the functionality can be done in the next iteration.

Disadvantages of Spiral Model:

- The spiral model is best suited for large projects only.
- The cost can be high as it might take a large number of iterations which can lead to high time to reach the final product.



$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

e^{+u_p}

ጵ

ጵጵጵ

888

5) Iterative Incremental Model

- The iterative incremental model divides the product into small chunks.
- Each iteration goes through the phases namely Requirement Analysis, Designing, Coding, and Testing.
- Detailed planning is not required in iterations.
- Once the iteration is completed, a product is verified and is delivered to the customer for their evaluation and feedback.

$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

ጵ

ጵጵጵ

888

e^{+up}

5) Iterative Incremental Model

Phases of IterativeIncremental Development Model:

- Inception phase
- Elaboration Phase
- Construction Phase
- Transition Phase

$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

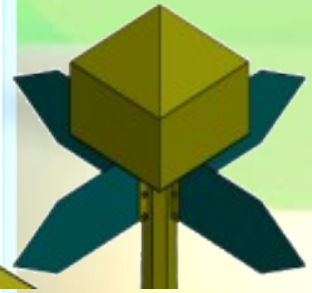
e^{+u_p}

ጵ

ጵጵጵ

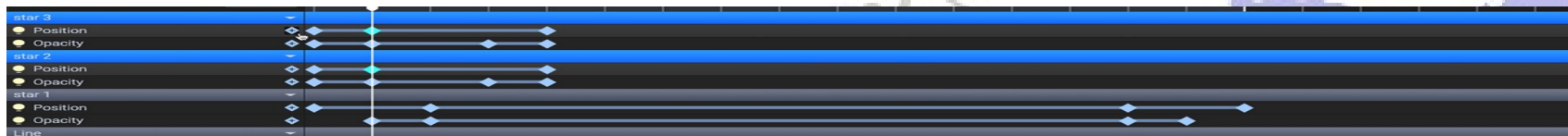
888

5) Iterative Incremental Model



Advantages of Iterative & Incremental Model:

- Any change in the requirement can be easily done and would not cost as there is a scope of incorporating the new requirement in the next iteration.
- Risk is analyzed & identified in the iterations.
- Defects are detected at an early stage.
- As the product is divided into smaller chunks it is easy to manage the product



$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

ጵ

ጵጵጵ

888

e^{+u_p}

5) Iterative Incremental Model

Disadvantages of Iterative & Incremental Model:

- Complete requirement and understanding of a product are required to break down and build incrementally.

$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

e^{+up}

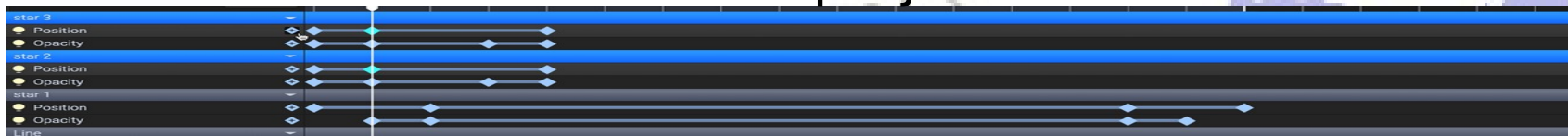
ጵ

ጵጵጵ

888

6) Big Bang Model

- Big Bang Model does not have any defined process.
- Money and efforts are put together as the input and output come as a developed product which might be or might not be the same as what the customer needs.
- The developer does the requirement analysis & coding and develops the product as per his understanding.
- This model is used for small projects only.
- There is no testing team and no formal testing is done, and this could be a cause for the failure of the project.



$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

ጵ

ጵጵጵ

888

e^{+u_p}

6) Big Bang Model

Advantages of Big Bang Model:

- It’s a very simple Model.
- Less Planning and scheduling is required.
- The developer has the flexibility to build the software of their own.

Disadvantages of the Big Bang Model:

- Big Bang models cannot be used for large, ongoing & complex projects.
- High risk and uncertainty.

$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

ጵ

ጵጵጵ

888

e^{+up}

7) Agile Model

- Agile Model is a combination of the Iterative and incremental model.
- This model focuses more on flexibility while developing a product rather than on the requirement.
- In Agile, a product is broken into small incremental builds.
- It is not developed as a complete product in one go.
- The next build is built on previous functionality.
- Each build increments in terms of features.
- In agile iterations are termed as sprints.
- Customer feedback is taken for improvement and his suggestions and enhancement are worked on in the next sprint.
- Testing is done in each sprint to minimize the risk of any failures.

$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

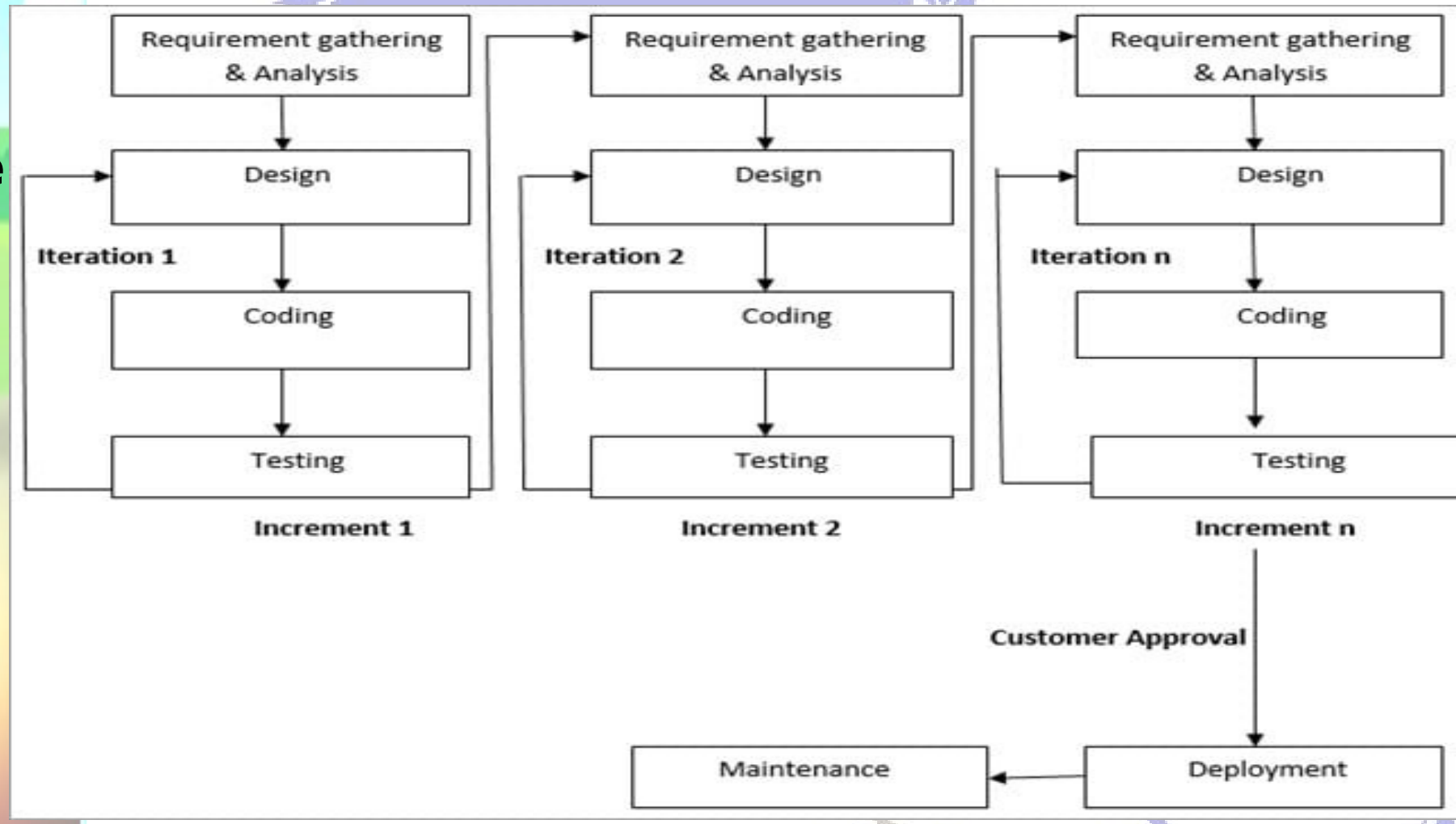
e^{+up}

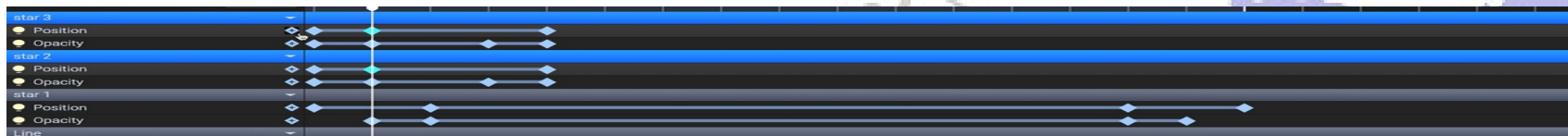
ጵ

ጵጵጵ

888

7) Agile Mode





$$f(e) = e_0 + \sum_{e=1} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

e^{+up}

ጵ

ጵጵጵ

888

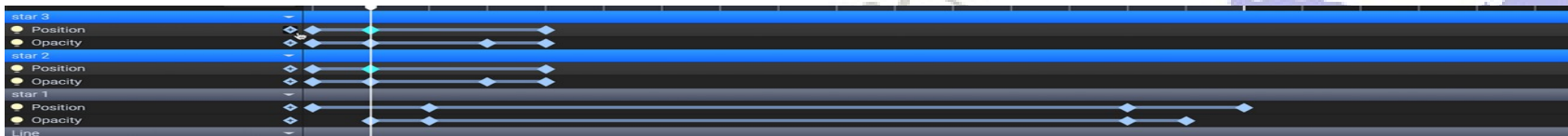
7) Agile Model

Advantages of Agile Model:

- It allows more flexibility to adapt to the changes.
- The new feature can be added easily.
- Customer satisfaction as the feedback and suggestions are taken at every stage.

Disadvantages:

- Lack of documentation.
- Agile needs experienced and highly skilled resources.
- If a customer is not clear about how exactly they want the product to be, then the project would fail.



$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

ጵ

ጵጵጵ

888

e^{+u_p}

Frontend vs Backend

- Frontend and Backend are the two most popular terms used in web development.
- The front end is what users see and interact with and the backend is how everything works.
- Each side needs to communicate and operate effectively with the other as a single unit to improve the website’s functionality.

$$f(e) = e_0 + \sum_{e=1} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

e^{+up}

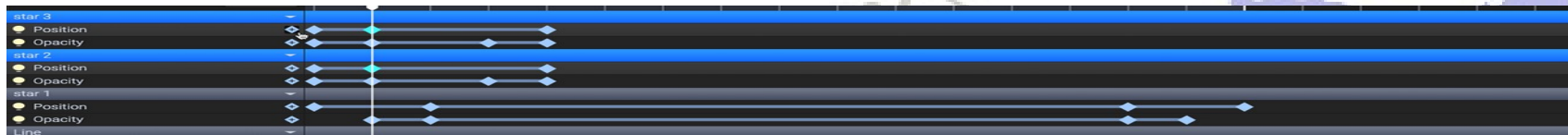
ጵ

ጵጵጵ

888

Frontend vs Backend

- The front end is the part includes the graphical user interface (GUI) and the command line including the design, navigating menus, texts, images, videos, etc. The backend, on the contrary, is part of the website users cannot see and interact with.
- Everything that happens in the background can be attributed to the backend.
-



$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢ ጽ ላሣ ጽ

ኢጽላሣ

EplusApp

e^{+up}

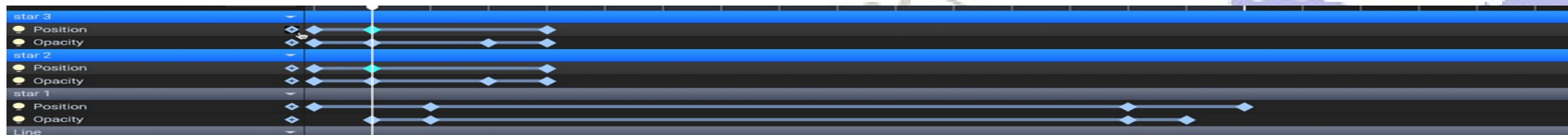
ጽ

ጽጽጽ

888

Front End Development

- The visual aspects of the website that can be seen and experienced by users are frontend. It is also referred to as the the development of a client side of the application.
- HTML,CSS,JAVASCRIPT,ANGULARjs,REACT,JQUERy,SASS,FLUTTER,



$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

e^{+up}

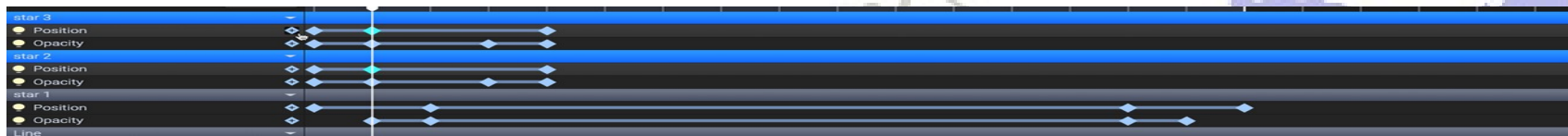
ጵ

ጵጵጵ

888

Front End Development

- **Responsiveness** and **performance** are the two main objectives of the **Front End**.
- The developer must ensure that the site is responsive i.e. it appears correctly on devices of all sizes no part of the website should behave abnormally irrespective of the size of the screen.



$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

e^{+u_p}

ጵ

ጵጵጵ

888

Back End Development

- The back-end portion is built by using some languages which are discussed below: PHP,C++,JAVA,Python,Node js,Express,Django,Ruby on Rails, Laravel ,Spring,C#, Ruby, GO, etc.

$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

e^{+u_p}

ጵ

ጵጵጵ

888

Ways to connect your front-end web code to the back-end

- An **application programming interface(API):** is mechanism that enable two software components to communicate with each other using a set of definitions and protocols.
- An API defines how a developer should request services from an operating system (OS) or other application, and expose data within different contexts and across multiple channels.

$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

e^{+up}

ጵ

ጵጵጵ

888 Ways to connect your front-end web code to the back-end

- There are four different types of APIs commonly used in web services: **public, partner, private and composite.**
- In the case of web and mobile frontends, the API is often based on HTTP request/response.

$$f(e) = e_0 + \sum_{e=1} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

e^{+up}

ጵ

ጵጵጵ

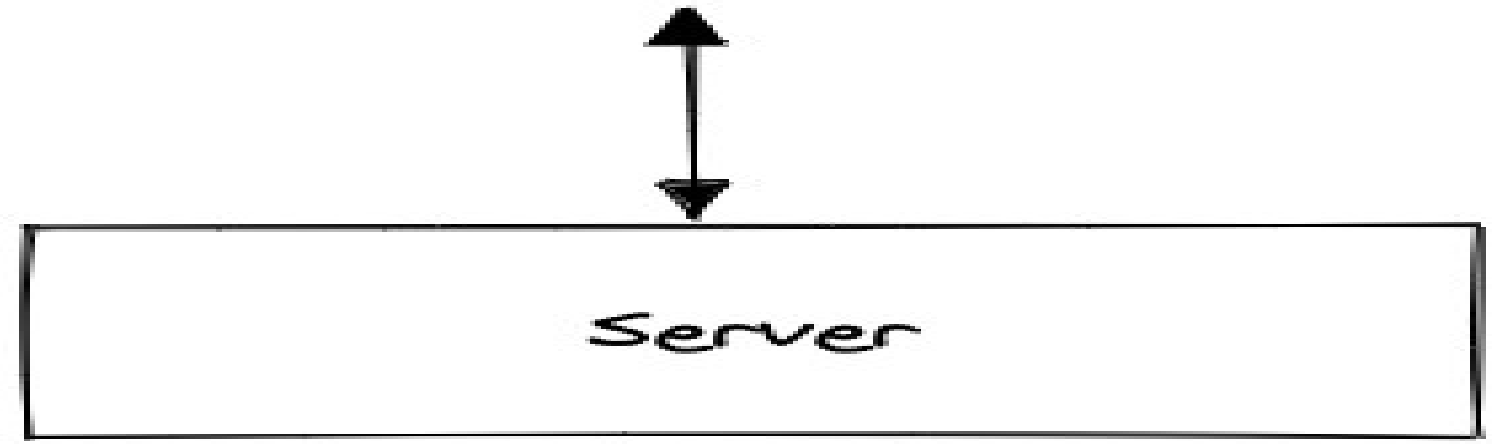
888

1)On The Server

- **On The Server- The** client-side code does not send API requests

  - The front-end code communicates with the back-end by making HTTP requests to the server's APIs which then processes the requests and returns responses to the front-end

  - The server provides everything - front and back.



$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

e^{+up}

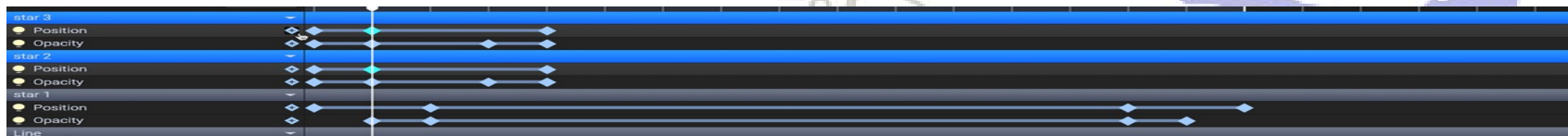
ጵ

ጵጵጵ

888

2)API Requests

- This is pretty popular because you can separate your client-side.
- The back-end code provides APIs that the client can use to interact with the server.
- The client-side code is responsible for sending API requests directly to the server's APIs to communicate with the back-end.
- All of your public content is kept separate from the things that you need to keep secret, or that need to be generated depending on the user. That stays on your server



$$f(e) = e_0 + \sum_{e=} e(Adress + Promotion + ecommerce)$$

ኢ ጽ ላሣ ጽ

ኢጽላሣ

EplusApp

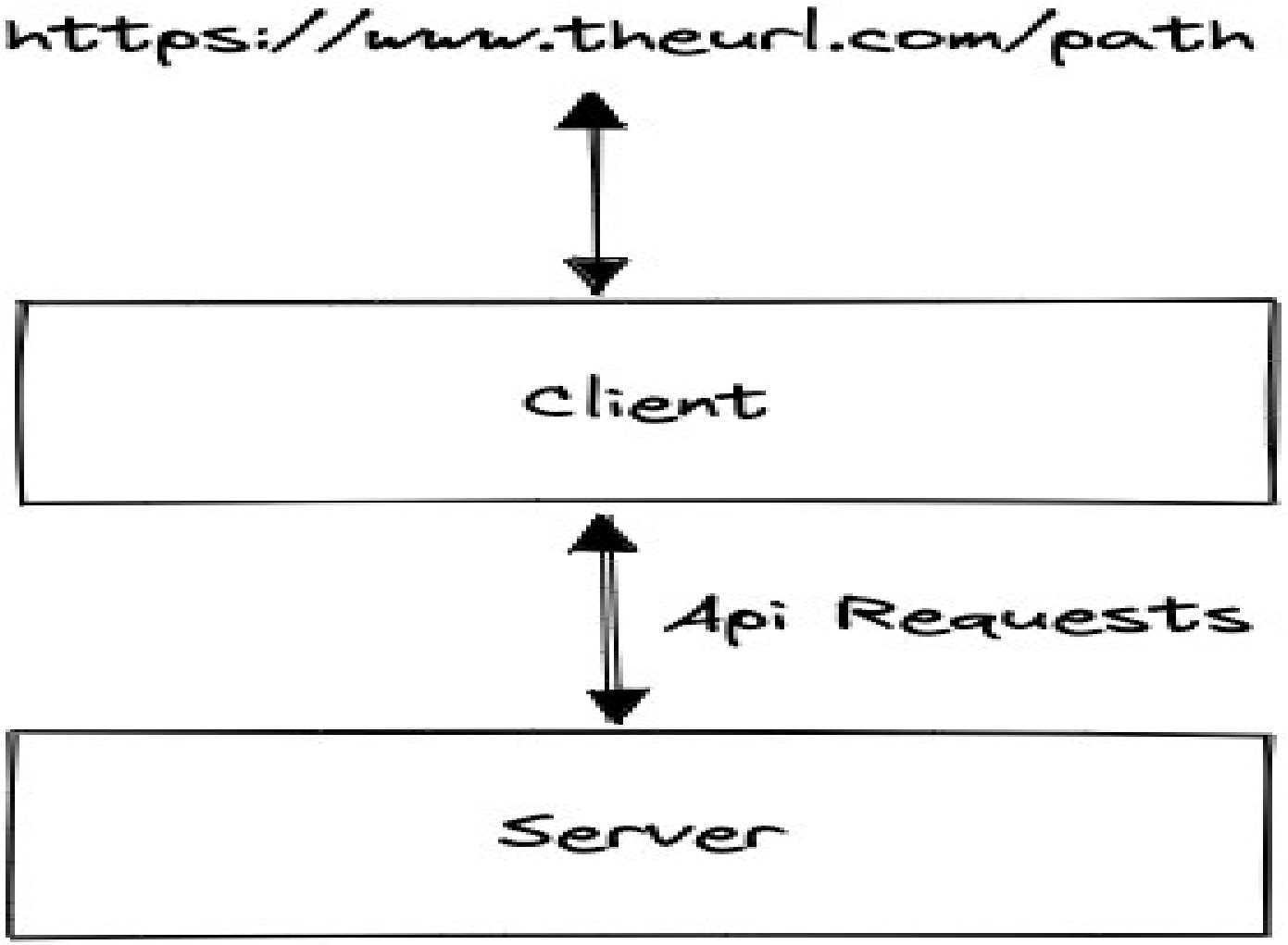
e^{+up}

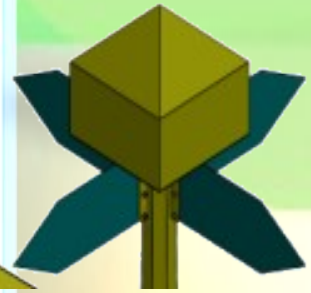
ጽ

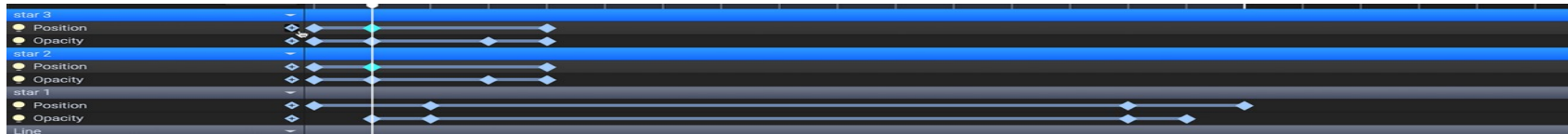
ጽጽጽ

888

2)API Requests







$$f(e) = e_0 + \sum_{e=1} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

e^{+up}

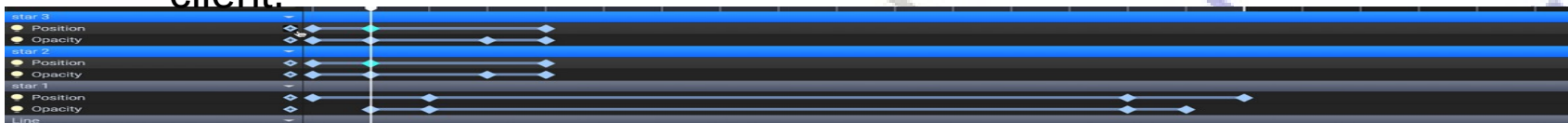
ጵ

ጵጵጵ

888

3. Serverless

- which then processes the requests and returns responses to the front-end
- You might not even need to run a server yourself, and that’s where serverless functions come in.
- This is kind of like API requests, but instead of those requests going to a URL on your server, they might go to a serverless function instead.
- The back-end code is responsible for processing the client's requests and generating responses, rather than exposing APIs directly to the client.



$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

e^{+up}

ጵ

ጵጵጵ

888

Software Quality Metrics

- **Software metrics can be classified into three categories :**

- **Product metrics** – Describes the characteristics of the product such as size, complexity, design features, performance, and quality level.

- **Process metrics** – These characteristics can be used to improve the development and maintenance activities of the software.

$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

e^{+up}

ጵ

ጵጵጵ

888

Software Quality Metrics

- **Project metrics** – This metrics describe the project characteristics and execution.
  - Eg. Including the number of software developers, the staffing pattern over the life cycle of the software, cost, schedule, and productivity.

$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

e^{+u_p}

ጵ

ጵጵጵ

888

Soft Ware Defects And Security Vulnerabilities

- **Software Defects**:-These are defects in the software or operating system of a computer or device, such as bugs, glitches, or crashes that can cause programs to malfunction or stop working altogether

- **Security vulnerabilities**: These are defects caused by security vulnerabilities in the device's software or operating system, which can be exploited by hackers or malicious software to gain access to the device or steal data.

$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

e^{+u_p}

ጵ

ጵጵጵ

888

Malicious software(malware)

- A type of software that is designed to harm or compromise a computer system or device without the user's knowledge .
- Malware is a broad category that includes a variety of different types of software, including viruses, trojans, ransomware, spyware, and adware.

$$f(e)=e_0+\sum_{e=1}^{\infty} e(Adress+Promotion+ecommerce)$$

ኢጵላሣ

EplusApp

e^{+u_p}

ጵ

ጵጵጵ

888

Types Malware?

1)Viruses: - A virus is a type of malware that can replicate itself and spread to other devices or systems.

- - Viruses can cause a range of problems, including data loss, system crashes, and performance issues.

2)Trojans: is a type of malware that disguises itself as legitimate software or files. Once installed, a trojan can perform a variety of malicious actions, including stealing personal information, creating back doors for remote access, and downloading additional malware.

$$f(e)=e_0+\sum_{e=1}^{\infty} e(Adress+Promotion+ecommerce)$$

ኢጵላሣ

EplusApp

ጵ

ጵጵጵ

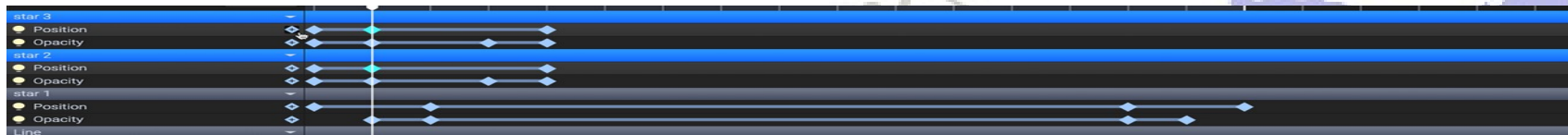
888

e^{+up}

Types Malware?

3)Ransomware: Ransomware is a type of malware that encrypts files on a user's device and demands payment in exchange for the decryption key. Ransomware attacks can be devastating for individuals and organizations, as they can cause data loss and financial damage.

4)Spyware: Spyware is a type of malware that is designed to monitor user activity and steal personal information. Spyware can be used to track online behaviour, steal login credentials, and collect sensitive information.



$$f(e) = e_0 + \sum_{e=1} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

e^{+up}

ጵ

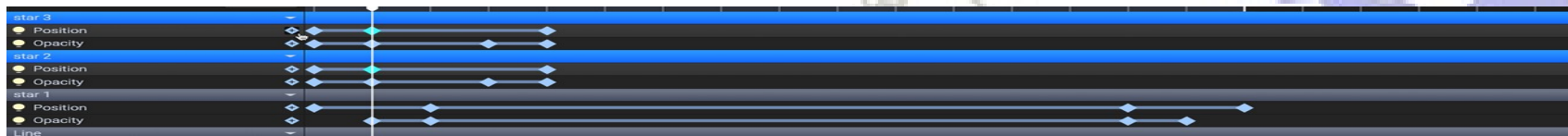
ጵጵጵ

888

Types Malware?

5)Adware: Is a type of malware that displays unwanted advertisements or pop-ups on a user's device. Adware can be used to generate revenue for attackers or to deliver additional malware to the device.

- Malware can affect a computer or other device through various means, including the file system, images, threads (processes), and other components.



$$f(e) = e_0 + \sum_{e=1} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

ጵ

ጵጵጵ

888

e^{+u_p}

Software Security

- **Cyber security** and **system security** are critical components of computer and information security
- **Cybersecurity** refers to the protection of computer systems, networks, and data from unauthorized access, theft, damage, or other malicious activities.
- **System security** refers to the measures taken to protect the hardware, software, and data within a specific computer system

$$f(e) = e_0 + \sum_{e=1} e(Adress + Promotion + ecommerce)$$

ኢ ጽ ላሣ ጽ

ኢጽላሣ

EplusApp

e^{+u_p}

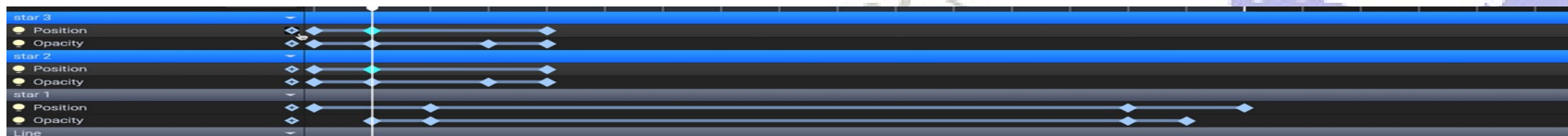
ጽ

ጽጽጽ

888

What is antivirus software?

- From a cyber security perspective, antivirus software can help protect against malware and other cyber threats that can compromise a system's security and steal sensitive data.
- From a system security perspective, antivirus software can also help protect against internal threats



$$f(e)=e_0+\sum_{e=1}^{\infty} e(Adress+Promotion+ecommerce)$$

ኢ ጽ ላሣ ጽ

ኢጽላሣ

EplusApp

e^{+up}

ጽ

ጽጽጽ

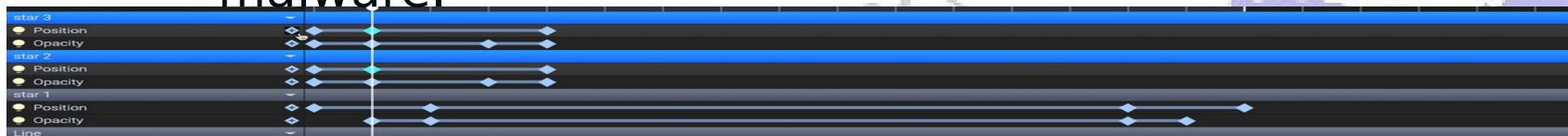
888

What is antivirus software?

- **Antivirus software:** is a program designed to prevent, detect, and remove malicious software. How it scan:

1)Signature-based scanning:comparing files on the computer to a database of known malware signatures. If a file matches a known signature, the antivirus software flags it as malware.

2)Heuristic analysis:This method involves identifying malware based on its behavior. Antivirus software can identify suspicious behaviour, such as attempting to modify system files or accessing sensitive information, and flag it as malware.



$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

e^{+u_p}

ጵ

ጵጵጵ

888

What is antivirus software?

3) Behavioural analysis: This method involves monitoring the behavior of software and processes running on the system to detect suspicious behaviour that could indicate malware.

4)Reputation-based scanning

5) Email scanning

6) Web-Scanning

7) Fileless malware detection

$$f(e) = e_0 + \sum_{e=1} e(Adress + Promotion + ecommerce)$$

ኢ ጽ ላሣ ጽ

ኢጽላሣ

EplusApp

e^{+u_p}

ጽ

ጽጽጽ

888

What programming language is used to develop antivirus?

Here is a list of some antivirus software and the programming languages they are developed in:

1) Norton Antivirus - Developed in C++ and Assembly language
2) McAfee Antivirus - Developed in C++ and Assembly language
3) Kaspersky Antivirus - Developed in C++ and Assembly language
4) Avast Antivirus - Developed in C and C++
5) AVG Antivirus - Developed in C++ and Assembly language
6) Bitdefender Antivirus -Developed in C++ and Assembly language
7) ESET Antivirus - Developed in C++ and Assembly language
8) Trend Micro Antivirus - Developed in C++ and Assembly language

$$f(e)=e_0+\sum_{e=1}^{\infty} e(Adress+Promotion+ecommerce)$$

ኢጵላሣ

EplusApp

e^{+u_p}

ጵ

ጵጵጵ

888

programming language is used to develop antivirus?

10)Smadav Antivirus-Developed Delphi programming language

11) Sophos Antivirus - Developed in C and C++

12) Malwarebytes Antivirus - Developed in C++

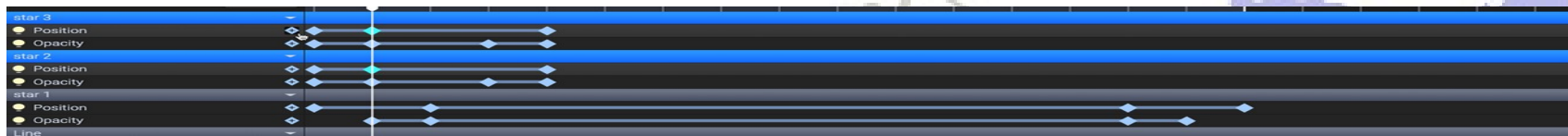
13) F-Secure Antivirus - Developed in C++ and Assembly language

14)Panda Antivirus - Developed in C++ and Assembly language

15) Comodo Antivirus - Developed in C++ and Assembly language

16)Windows Defender (built-in to Windows operating system) - Developed in C++

17) Malwarebytes -Developed in java and C#



$$f(e)=e_0+\sum_{e=1}^{\infty} e(Adress+Promotion+ecommerce)$$

ኢጵላሣ

EplusApp

ጵ

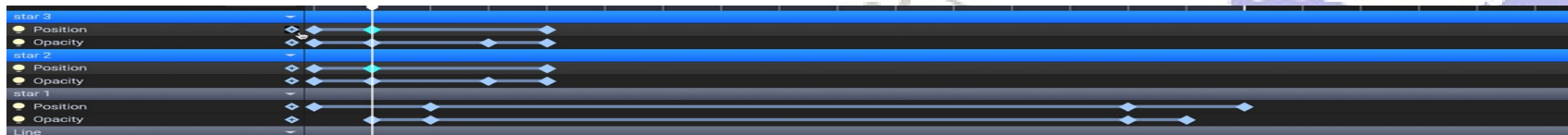
ጵጵጵ

888

e^{+u_p}

programming language used to develop antivirus

- **C++** :Is a popular language for developing antivirus software due to its speed, performance, and low-level access to system resources.
  - It allows for direct manipulation of system resources
  - C++ also has access to a wide range of libraries, such as the ClamAV library.
  - This makes it easier to develop antivirus software.



$$f(e)=e_0+\sum_{e=1}^{\infty} e(Adress+Promotion+ecommerce)$$

ኢጵላሣ

EplusApp

e^{+u_p}

ጵ

ጵጵጵ

888

programming language used to develop antivirus

- However, developing antivirus software in C++ can be challenging
  - requires a strong understanding of low-level programming concepts
  - requires a strong understanding of security vulnerabilities,and operating system internals
  - It also requires a significant amount of time and effort.
- **C#:** is a high-level programming language that is widely used for developing Windows applications, including antivirus software.

$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

ጵ

ጵጵጵ

888

e^{+u_p}

programming language used to develop antivirus

- C# provides a number of advantages for antivirus software development, including:
  - Access to system resources
  - .NET framework
  - Object-oriented programming
  - Windows integration
- Limitation: not be as performant as C++
- not portable like python and java

$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

ጵ

ጵጵጵ

888

e^{+u_p}

programming language used to develop antivirus

- **Java:**itself is a powerful programming language with many features and capabilities, and it can be used to develop antivirus software
- However, there are some limitations to using Java for this purpose:
  - **Performance**
  - **Access to system resources**
  - **Vulnerabilities**
  - **Compatibility**

$$f(e)=e_0+\sum_{e=1}^{\infty} e(Adress+Promotion+ecommerce)$$

ኢጵላሣ

EplusApp

e^{+u_p}

ጵ

ጵጵጵ

888

programming language used to develop antivirus

- **Python:** it is possible to develop antivirus software using Python.

  Python has several libraries and modules that can be used to build antivirus software, such as the os, subprocess, and hashlib libraries.

- **Limitations:**
  - **Performance**
  - **Access to system resources**
  - **Vulnerabilities**
  - **Compatibility**

$$f(e) = e_0 + \sum_{e=1} e(Adress + Promotion + ecommerce)$$

ኢ ጽ ላሣ ጽ

ኢጽላሣ

EplusApp

e^{+u_p}

ጽ

ጽጽጽ

888

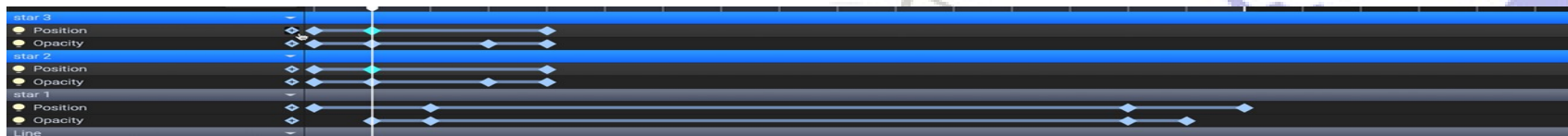
programming language used to develop antivirus

- many antivirus software products use a combination of programming languages
- C++ and Python together develop antivirus software that is:

  - fast and efficient at low-level system programming tasks

  - flexible and easy to use for high-level scripting and automation tasks

- **ClamAV** - The core engine is written in C++, while the user interface and scripting functionality are implemented in Python.



$$f(e) = e_0 + \sum_{e=1}^{\infty} e(Adress + Promotion + ecommerce)$$

ኢጵላሣ

EplusApp

ጵ

ጵጵጵ

888

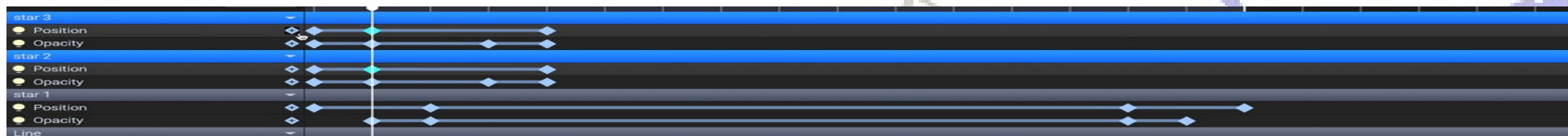
e^{+u_p}

programming language used to develop antivirus

- **Avast**:- C++ is used for the low-level components

  - Python is used for the higher-level components

- **McAfee,Bitdefender**



$$f(e)=e_0+\sum_{e=1} e(Adress+Promotion+ecommerce)$$

ኢጵላሣ

EplusApp

e^{+u_p}

ጵ

ጵጵጵ

888

Antivirus corresponds to malware

Antivirus software corresponds to the types of malware they can scan and remove:

- Norton Antivirus - can scan and remove viruses, spyware, and other types of malware.
- McAfee Antivirus - can scan and remove viruses, spyware, trojans, worms, and other types of malware.
- Kaspersky Antivirus - can scan and remove viruses, spyware, trojans, worms, ransomware, and other types of malware.
- Avast Antivirus - can scan and remove viruses, spyware, trojans, worms, ransomware, and other types of malware.
-

$$f(e)=e_0+\sum_{e=1}^{\infty} e(Adress+Promotion+ecommerce)$$

ኢጵላሣ

EplusApp

e^{+up}

ጵ

ጵጵጵ

888

Antivirus corresponds to malware

- Bitdefender Antivirus - can scan and remove viruses, spyware, trojans, worms, ransomware, and other types of malware.
- AVG Antivirus - can scan and remove viruses, spyware, trojans, worms, ransomware, and other types of malware.
- Avira Antivirus - can scan and remove viruses, spyware, trojans, worms, ransomware, and other types of malware.
- Trend Micro Antivirus - can scan and remove viruses, spyware, trojans, worms, ransomware, and other types of malware.
- ESET Antivirus - can scan and remove viruses, spyware, trojans, worms, ransomware, and other types of malware.
- These anti-viruses can scan and remove a wide range of malware types, but no antivirus software is 100% effective at detecting and removing all malware.

$$f(e) = e_0 + \sum_{e=1}^{\infty} e\left(Adress + Promotion + ecommerce\right)$$

ኢ ጽ
ላ ፣ ጽ

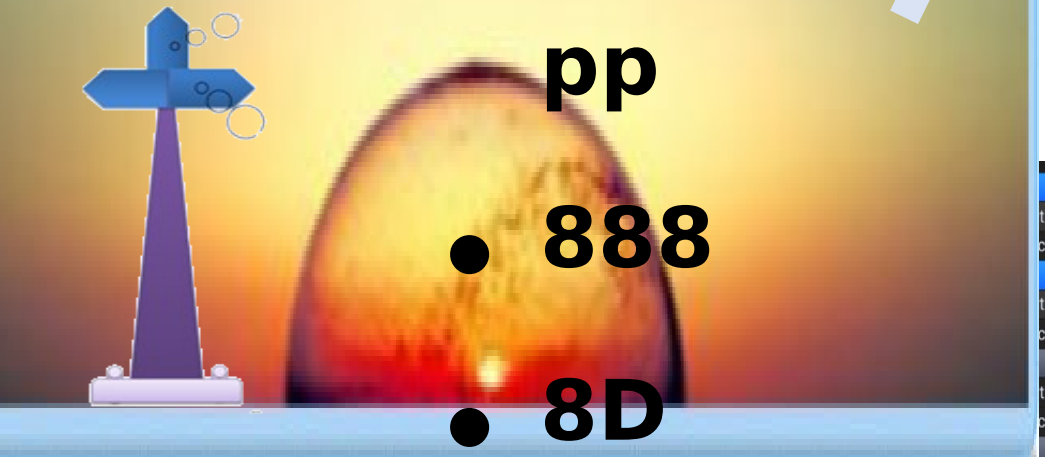
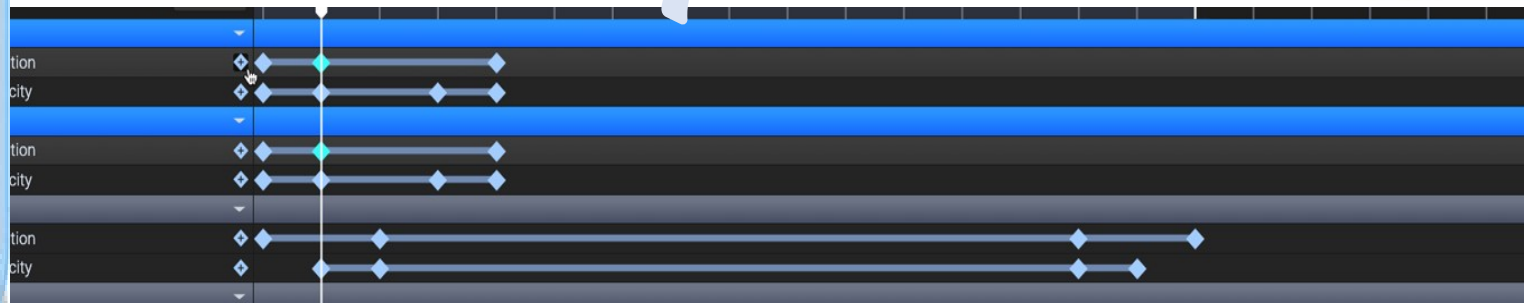
ኢጽላ፣ e^{+u_p}

EplusApp

ጽ ጽ ጽ

888

- ኢጽላ፣ጽ
- EplusA
- pp
- 888
- 8D



እናመሰግናለን!